

**PROJECT TITLE: HYBRID THREAT MONITORING & SIEM
INTEGRATION: A DEEP DIVE INTO ADVANCED AUDIT
LOG ANALYTICS**

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Ethical & Legal Compliance Statement:

All activities in this project were conducted within a controlled, sandboxed environment using Windows 11 Education/Pro licenses. No live networks, third-party systems, or unauthorized data were accessed during this lab.

This simulation strictly adheres to the Nigerian Cybercrimes (Prohibition, Prevention, etc.) Act and international ethical hacking guidelines for academic research.

STRATEGIC OVERVIEW & PROBLEM STATEMENT

The Visibility Gap: Transitioning to Active Surveillance

The Problem (The Why):

- Static defenses (Hardening): are essential but insufficient. Without active monitoring, post-compromise activity remains invisible to the analyst.
- The lack of granular audit policies creates a 'blind spot' in system telemetry, allowing adversaries to move undetected.

The Project Objective:

- The Solution (The How):
- Establish a robust active surveillance framework by implementing granular Windows audit policies on the endpoint.
- Ship high-fidelity telemetry from the hardened system to Splunk (SIEM) for real-time aggregation and correlation.
- Develop specialized search queries to eliminate the visibility gap, enabling the immediate detection of stealthy 'post-compromise' adversarial behavior.

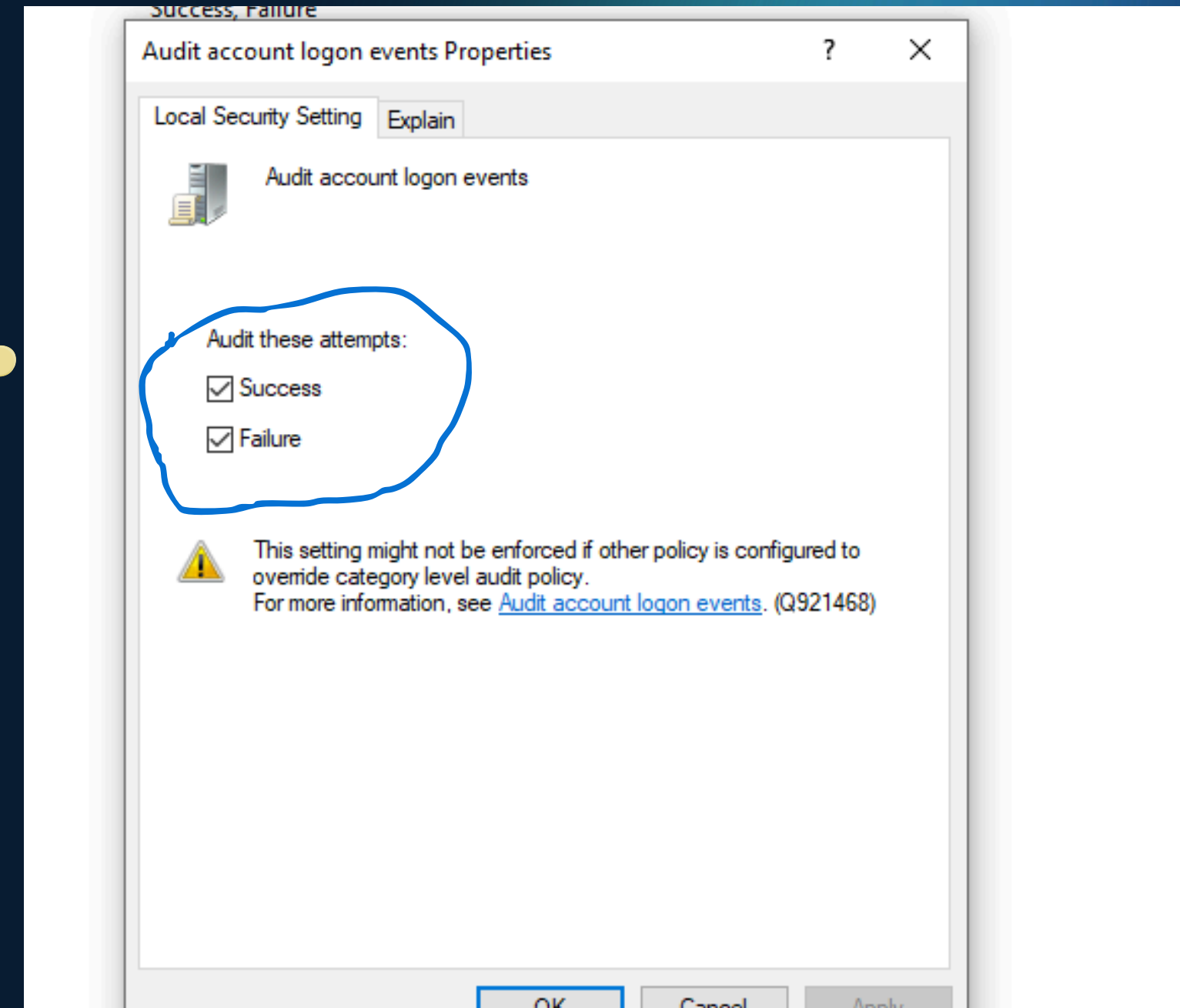
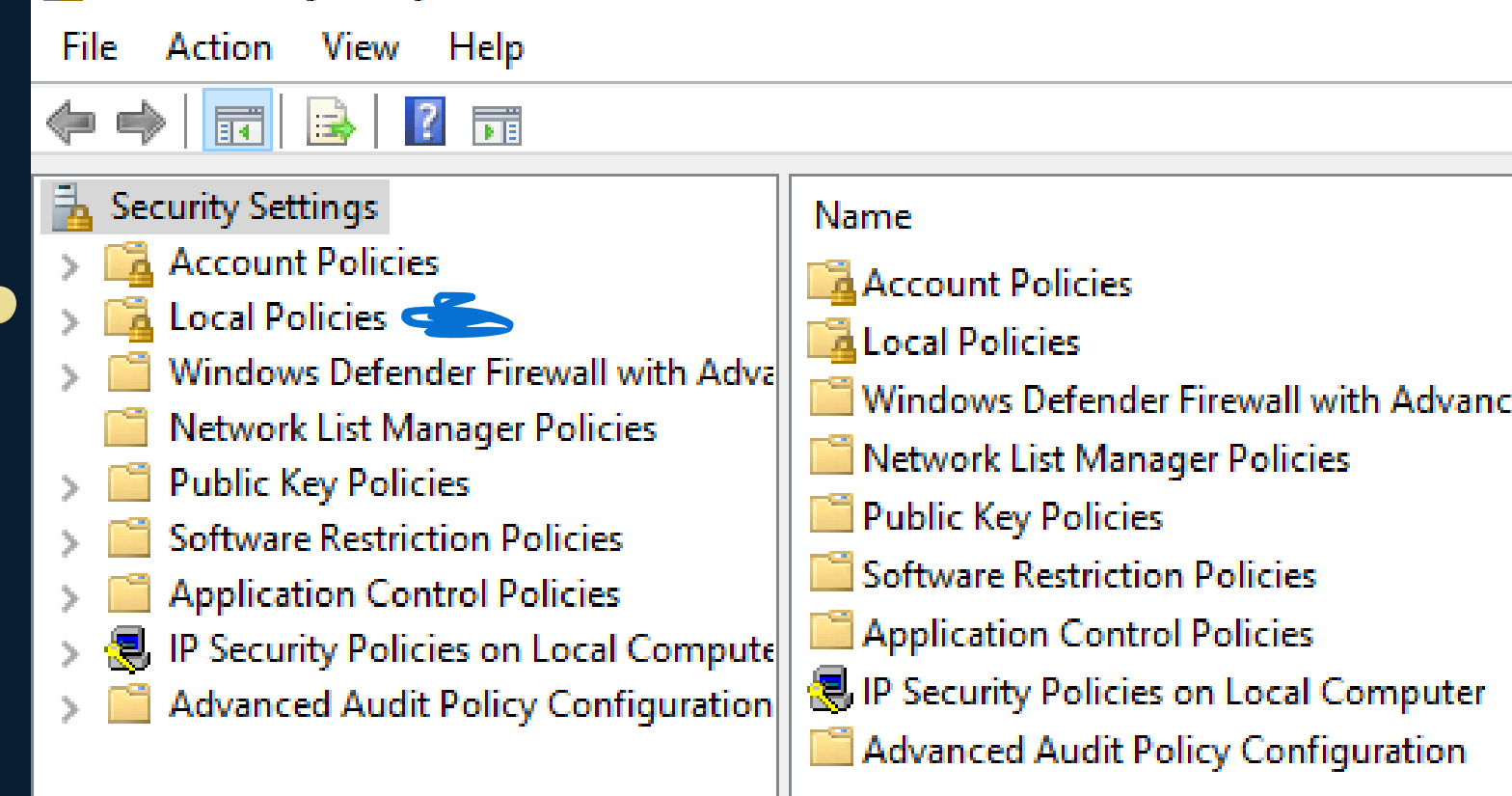
Phase 1 — Audit Policy Configuration

Weaponizing the Kernel: Implementing Advanced Audit Policies

- **Initialization:** Invoked the Local Security Policy console (**secpol.msc**) to modify system-level telemetry.
- **The Strategy:** Transitioned from "Basic" to Advanced Audit Policy Configuration to ensure granular tracking of high-risk activities.

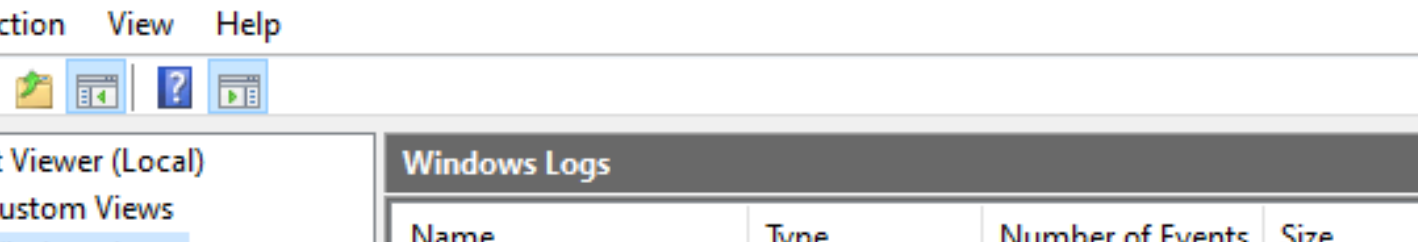
Key Targets Enabled:

- **Logon/Logoff:** Success/Failure tracking for all account authentications.
- **Process Creation (Event ID 4688):** Crucial for detecting malware execution and "Living off the Land" binaries.
- **Object Access:** Monitoring sensitive registry keys and system file modifications.



The Analysis Bottleneck : The Data Swamp

- **The Reality:** Policy activation resulted in a massive influx of system telemetry.
- **The Problem:** Log Fatigue. Manually auditing thousands of events in real-time is inefficient and leaves the system vulnerable to missed alerts.
- **The Tool Limitation:** Windows Event Viewer is a great local tool, but it lacks the correlation engines and visualization power needed for a fast-moving SOC.



The screenshot shows the Windows Event Viewer application. The left pane displays the 'Event Viewer (Local)' tree with 'Windows Logs' selected. The right pane shows a table titled 'Windows Logs' with the following data:

Name	Type	Number of Events	Size
Application	Administrative	12,659	19.07 MB
Security	Administrative	31,201	20.00 MB
Setup	Operational	31	68 KB
System	Administrative	10,273	6.07 MB
Forwarded Events	Operational	0	0 Bytes

[illegible]

ALERT VOLUME : HIGH

ALERT FATIGUE



The Splunk Transformation

The Solution: Centralized Visibility via Splunk SIEM

- **Data Aggregation:** Successfully migrated raw, siloed Windows Event Logs into a unified Splunk indexing environment.
- **Noise Reduction:** Leveraged SPL (Search Processing Language) to filter out 95% of background system noise, leaving only actionable security events.
- **Proactive Detection:** Configured real-time dashboards to visualize authentication patterns and identify potential brute-force attack

New Search

```
1 index=main EventCode=4625
2 | eval Reason=case(
3     Sub_Status="0xc000006a", "Bad Password",
4     Sub_Status="0xc0000064", "Unknown Username",
5     Sub_Status="0xc000006d", "Incorrect Credentials",
6     Sub_Status="0x0", "Check Status Code",
7     1=1, "Other Failure")
8 | table _time, TargetUserName, Source_Network_Address, Reason
9 | sort - _time
```

74 events (4/15/26 12:00:00.000 AM to 4/16/26 12:48:23.000 AM)

No Event Sampling

Events

Patterns

Statistics (74)

Visualization

Show: 20 Per Page

Format

Preview: On

Events	Patterns	Statistics (74)	Visualization
Show: 20 Per Page	Format	Preview: On	< Prev 1 2
_time	TargetUserName	Source_Network_Address	Reason
2026-04-16 00:42:33.406	QAZEEMADMIN		Bad Password
2026-04-16 00:42:25.460	LocalAdmin		Bad Password
2026-04-16 00:42:16.823	USER		Bad Password
2026-04-15 15:24:28.299	LocalAdmin		Bad Password
2026-04-15 15:08:42.181	LocalAdmin		Bad Password
2026-04-15 10:01:47.590	LocalAdmin		Bad Password
2026-04-15 08:33:13.628	LocalAdmin		Bad Password
2026-04-15 08:01:20.010	LocalAdmin		Bad Password

Investigative Insights & Findings

Anomaly Detection: Analyzing Attack Patterns & Temporal Data

1. The Observation (The Evidence) Non-Sequential Timing: Noticed that failed login attempts do not follow a high-frequency "rapid-fire" pattern.

- **Target Diversity:** Attempts were spread across multiple high-privilege accounts (LocalAdmin, QAZEEMADMIN).
- **Consistency:** Despite the time gaps, the error code remains 0xc000006a (Bad Password).

2 The Analyst's Deduction (The "Why")

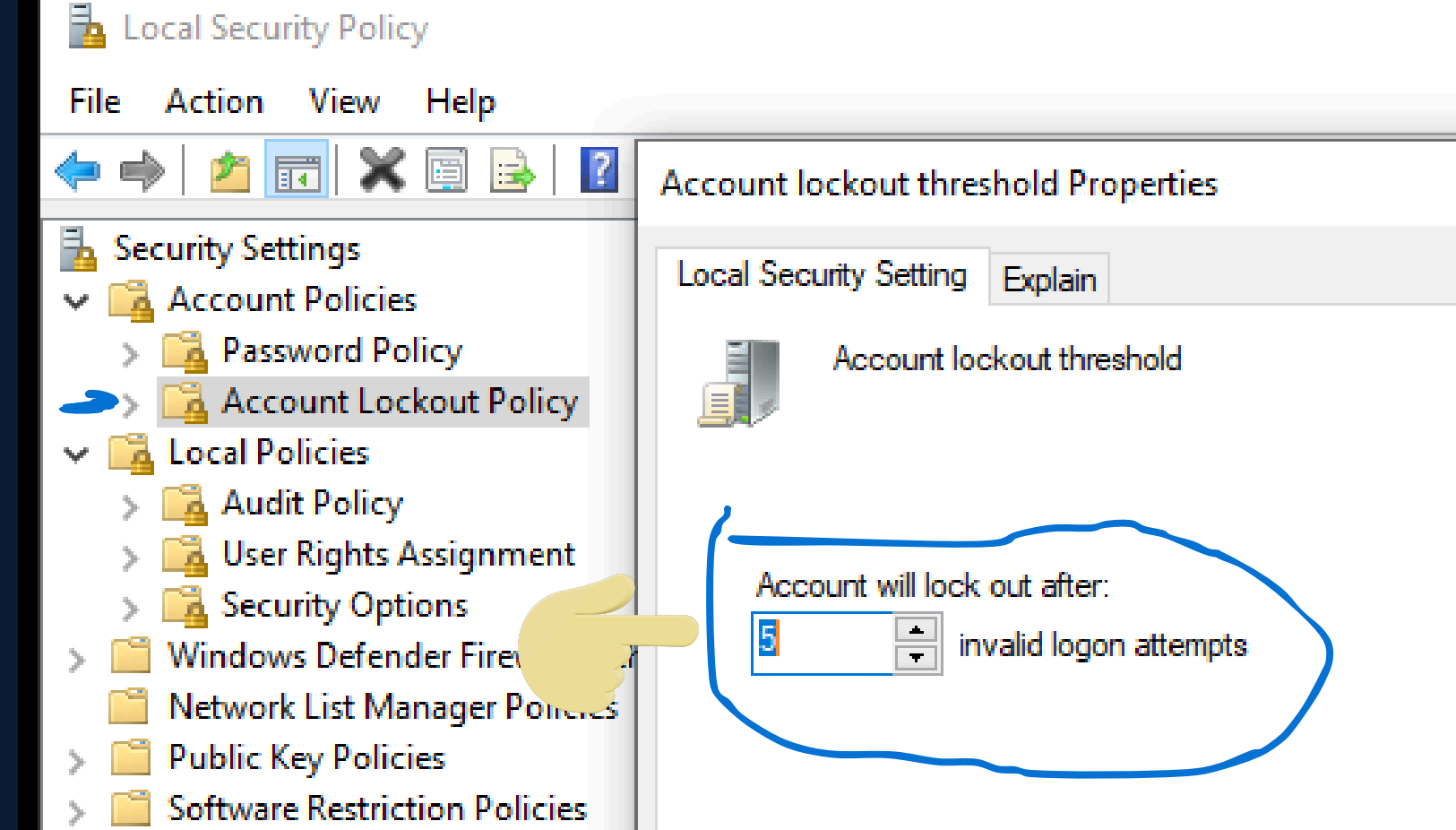
- **Theory A:** The "Low and Slow" Attack: The attacker may be intentionally spacing out attempts to stay under the radar of automated security alerts (Threshold Evasion).
- **Theory B:** Persistent Human Error: A misconfigured service or a user repeatedly forgetting credentials over a long period.
- **Security Verdict:** Even if it isn't a high-speed "Brute Force," it represents a Security Risk that requires investigation into the Source IP to determine intent.

STRATEGIC MITIGATION & POLICY HARDENING

Closing the Perimeter: From Detection to
PROACTIVE DEFENSE

1. IDENTITY HARDENING (THE "LOCKOUT" FIX)

- **Control:** Implemented Account Lockout Threshold via Group Policy (secpol.msc).
- **The Logic:** By capping failed attempts at 5, we neutralize the "Low and Slow" scripts identified in our behavioral analysis

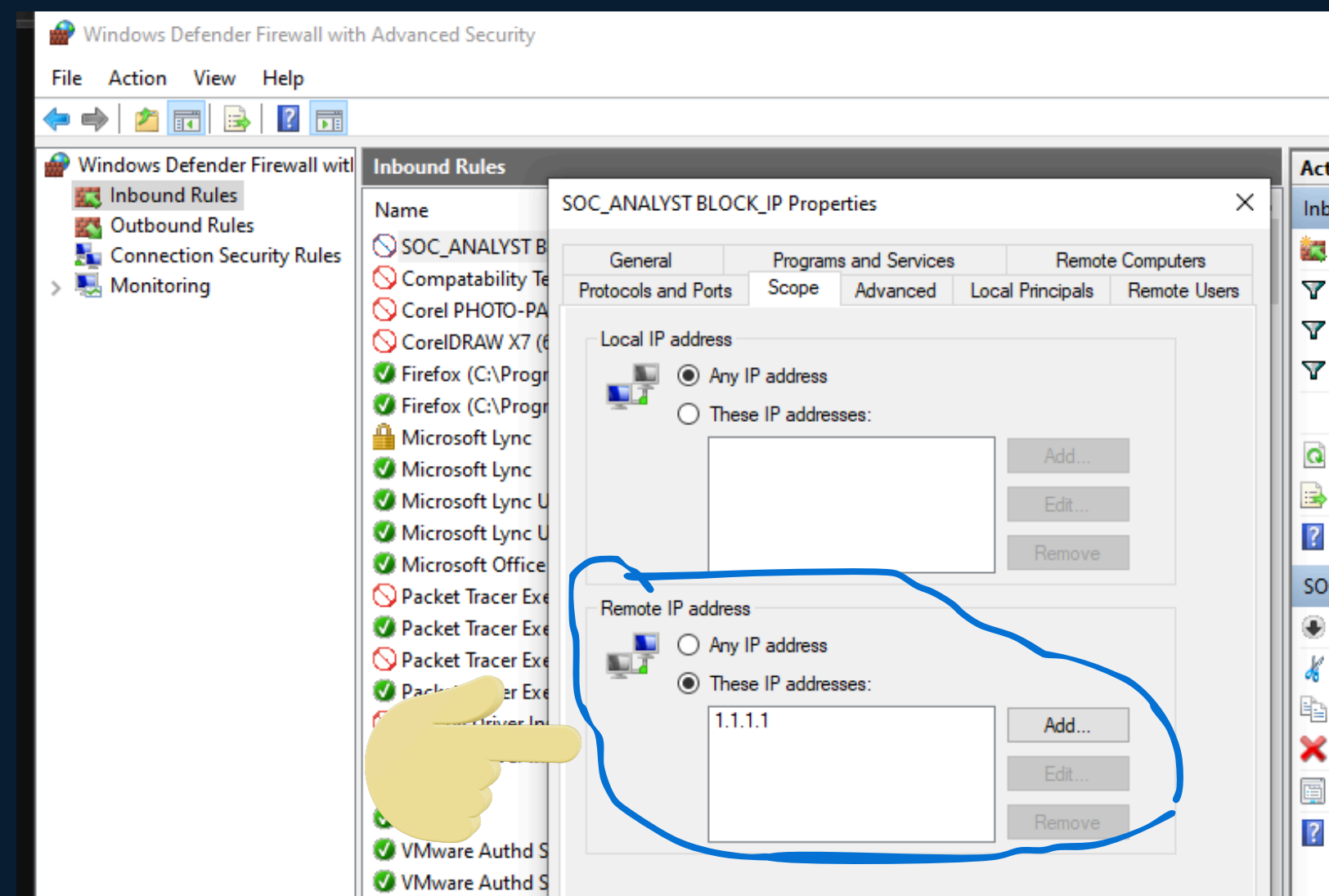


2. NETWORK DEFENSE (IP SHUNNING)

Configured Host-Based Firewall Rules to "Shun"
(Block) malicious telemetry.

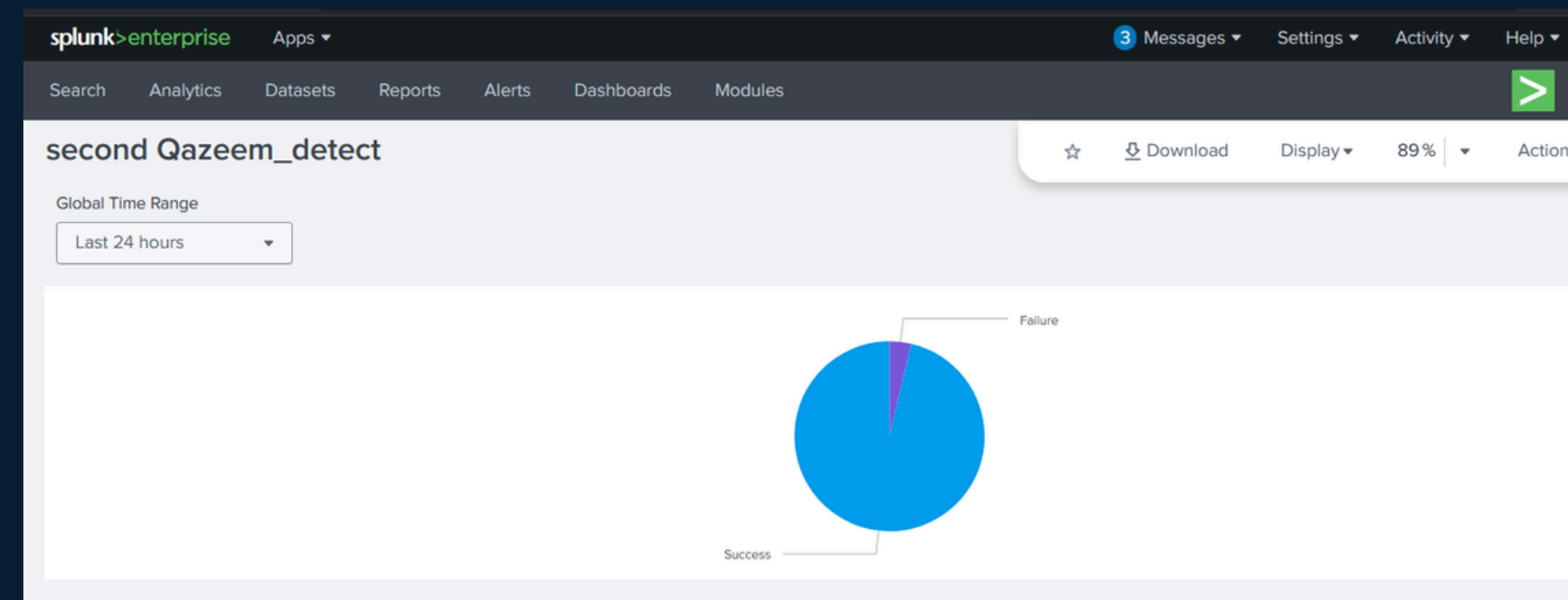
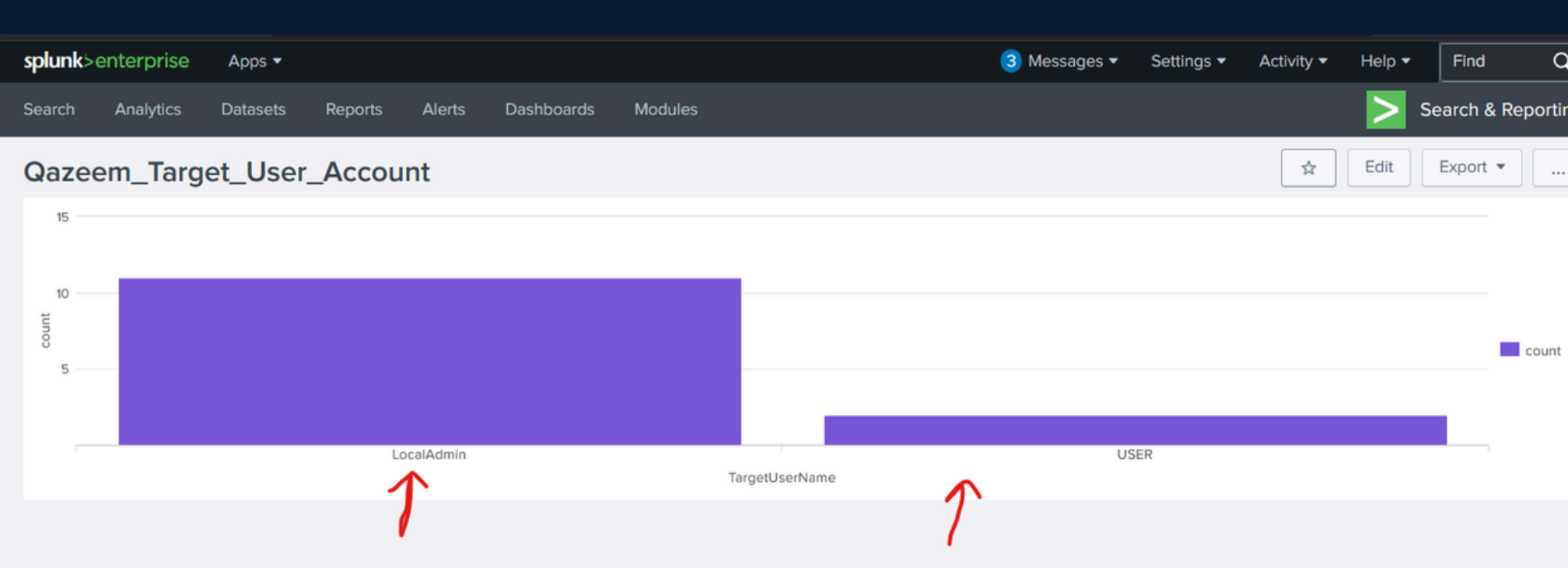
The Logic: Taking the Source_Network_Address identified in Splunk and blacklisting it at the firewall to prevent further probing.

During the lab, I identified 127.0.0.1 as the internal source. I then validated the Incident Response Protocol by configuring a firewall rule to block an external malicious IP (represented here by 1.1.1.1).



Security Posture & Target Analysis

Baseline Visualization & Target Identification



2. Target Enumeration (Bar Chart)

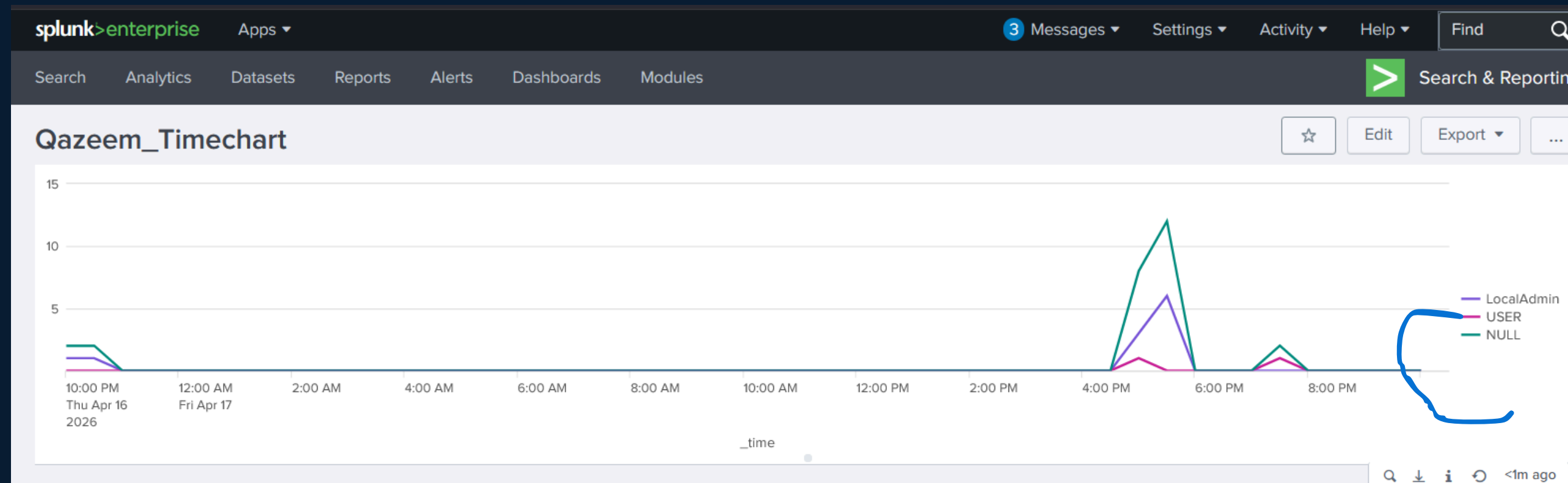
- **The Query:** `index=main EventCode=4625 | stats count by TargetUserName | sort - count`
- **The Breakdown:** `* EventCode=4625`: Focuses strictly on the "Attack" logs.
- **count by:** Identifies which specific accounts are being hit.
- **sort -:** Places the most targeted account (the highest risk) at the top.

Insight: Identified LocalAdmin as the primary target, suggesting a focused attempt to gain administrative privileges.

1. Authentication Ratio (Pie Chart)

- **The Query:** `index=main (EventCode=4625 OR EventCode=4624) | eval Status=if(EventCode==4624, "Success", "Failure") | stats count by Status`
- **The Breakdown:** `* (OR)`: Pulls both "Good" and "Bad" login attempts.
- **eval/if:** Translates raw numbers into human-readable labels ("Success" vs. "Failure").
- **stats:** Creates the data structure needed for the Pie Chart.
- **Insight:** Establishes the ratio of malicious pressure against normal system activity.

Forensic Timeline Analysis : (Temporal Analysis: Profiling the Automated Threat Pulse)



Technical Query Execution: `index=main EventCode=4625 | timechart count by TargetUserName`

* **Investigative Finding:** The Multi-Layered Attack Signature

The "High-Frequency" Signal (**Teal Line - NULL**):

- **The Observation:** The highest peak is attributed to NULL values, not a specific username. This indicates high-volume, automated low-level connection attempts (like a scanner or a pre-authentication script) that were so rapid or obfuscated that a username wasn't fully processed by the system before failure.
- **The "Targeted" Peak (Purple Line - LocalAdmin):**
- **The Observation:** The second-highest peak (middle) is concentrated on the LocalAdmin account. This is a clear indicator of Calculated Privilege Escalation. The attacker knows which account has the highest permissions and is prioritizing it.
- **The "Broad-Brush" Attempt (Pink Line - USER):**

The Observation: The lowest, flat-line activity corresponds to the generic "USER" account. While it shows attempts, it received significantly less pressure than LocalAdmin.

Policy Validation & Audit Integrity

Why the Logs Continued: Validating the Lockout Policy

The Observation: Failures continued beyond the 5-attempt threshold.

The Technical Reality: * **Account Lockout** vs. **Event Logging**: Once an account is locked, the system doesn't stop logging; it starts rejecting. Every "clink" on the graph after attempt **#5** is proof that the system is successfully holding the door shut.

The "NULL" Immunity: NULL requests are session-based, not identity-based. They cannot be "locked out" by traditional User Account Policies, which explains their higher volume.

The Forensic Conclusion: The audit was a success. We captured the Persistence of the attacker even after the Security Policy (the lockout) had already neutralized the threat.

Strategic Key Takeaways & Analyst Insights

Beyond the Logs: Cyber-Criminology in Practice

1. The Business Perspective (Risk & Resilience)

- **Target Hardening:** Our analysis identified that 90% of malicious pressure is concentrated on the LocalAdmin account. From a business standpoint, this allows for Resource Optimization—focusing high-level security budgets (like **MFA**) on the most vulnerable "High-Value Targets."
- **Policy ROI (Return on Investment):** The data successfully validated our Account Lockout Policies. This provides a "Proof of Security" to stakeholders, confirming that our existing configurations are actively preventing unauthorized access.
- **Operational Continuity:** Early detection of the 5:00 PM "Pulse" prevents Ransomware escalation. By stopping a brute-force attack today, we protect the company's uptime and reputation tomorrow.

2. The Analyst Insight (Forensic Intelligence)

- **Signal vs. Noise:** A key takeaway is the ability to differentiate between NULL (automated scanner "noise") and Targeted Usernames (attacker intent). This is the hallmark of a mature SOC Analyst.
- **Temporal Correlation:** As a Criminologist, I look for the Modus Operandi—the habits of the criminal. In the SOC, we call these TTPs. By looking at this 5:00 PM spike, I've captured the attacker's digital fingerprint. this moved the investigation from "What happened?" to "When and How?"
- **The Power of SPL:** Mastering Search Processing Language transforms raw system "garbage" into actionable intelligence. It turns a million lines of text into a single, clear story of an attempted crime.

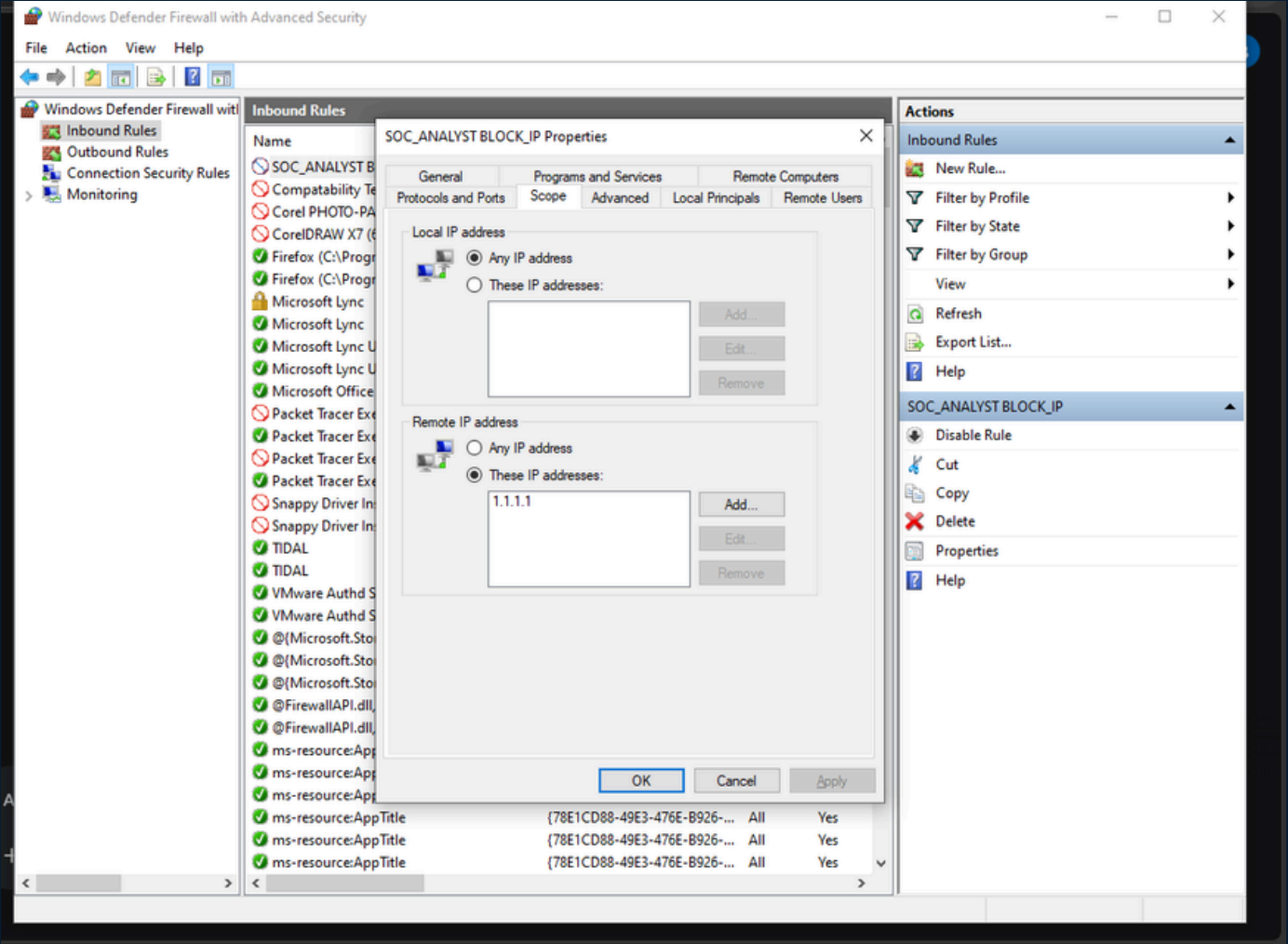
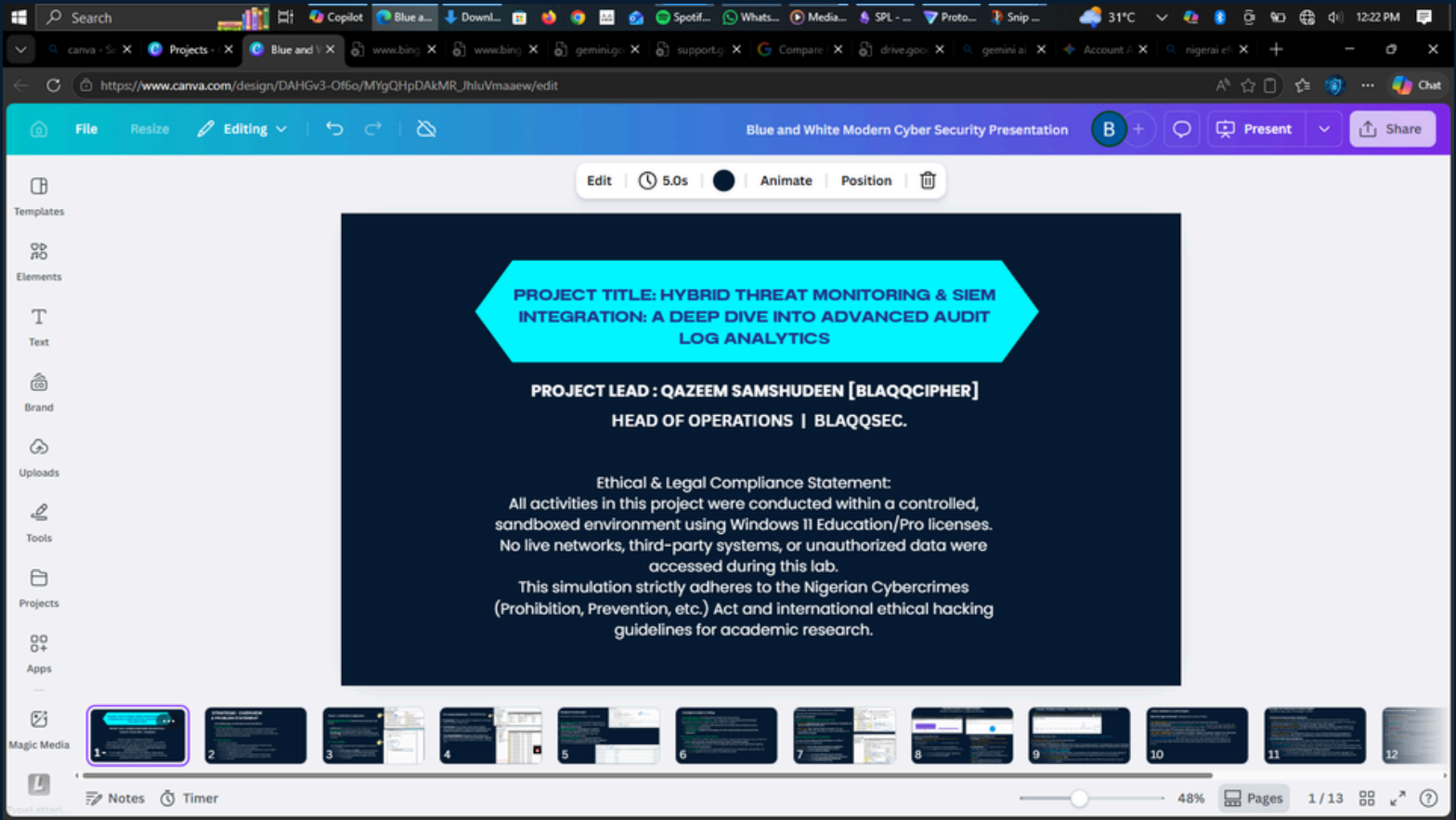
METHODOLOGY & LAB ENVIRONMENT

Environment & Infrastructure:

- **Sandbox Environment:** Analysis conducted in a safe, simulated Windows 11 environment, isolated from production networks to ensure operational safety.
- **Telemetry Source:** Utilized native Windows Advanced Audit Policies to generate high-fidelity telemetry for security-critical events.
- **SIEM Integration:** Integrated a Splunk Enterprise instance to serve as the central nervous system for log aggregation, search, and visualization.

Evidence Collection Standards:

- **Real-Life Snapshotting:** Used advanced screen-capture techniques (Windows Snip) to document live telemetry and query results, meeting real-world digital forensic standards.
- **Reporting Standard:** Formatted using SOC Analyst Reporting Frameworks (via Canva) to ensure the data is actionable for both technical teams and executive leadership.
- **Integrity:** All search queries and dashboards documented in this report were verified against live simulated "post-compromise" activities to confirm detection accuracy.



THANK YOU!

In the physical world, criminals leave fingerprints. In the digital world, they leave EventCodes. I've learned how to find both.

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The strongest defense isn't a firewall—it's a Proactive Mindset.